

UNC1549 — Smoke Sandstorm

Threat Actor Profile | Aviation & Aerospace

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AI-ASSISTED REPORT — IMPORTANT DISCLOSURE

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Attribution Warning: Nation-state attribution is probabilistic. Confidence levels reflect analytical assessment only — not proof.

1. At a Glance

Attribute	Value
Primary Alias	UNC1549
MITRE ATT&CK Group	Not formally assigned; tracked as UNC1549 by Mandiant/Google
Nation-State Sponsor	Iran — IRGC-linked (assessed High confidence)
Motivation	Espionage: aerospace/defence intelligence; targeting of Israeli and Gulf military capabilities
Active Since	June 2022 (confirmed); surge from late 2023
Activity Status	Active — elevated tempo through 2025–2026
Sophistication	Advanced — custom toolchain; cloud-native C2 tradecraft
Primary Sectors	Aerospace, Defence, Aviation supply chain, Government (defence-adjacent)
Primary Geographies	Israel, UAE, Turkey, India, Albania; Western Europe from Sep 2025 (Portugal, Sweden, Denmark)
Assessment Confidence	High — Mandiant/Google primary source; independent corroboration

2. Identity & Aliases

Vendor	Designation
Mandiant / Google Cloud	UNC1549
Microsoft	Smoke Sandstorm (formerly BOHRIUM)
Picus Security / Community	Nimbus Manticore
MITRE ATT&CK	Not assigned (campaign-level reporting only)
CISA	Referenced in Iran-nexus advisories

3. Attribution Assessment

Assessment: Iran — IRGC-linked. Confidence: High.

UNC1549 is assessed as Iranian state-directed based on: targeting exclusively aligned with Iranian strategic intelligence requirements (Israeli and Gulf military aviation/aerospace capability); use of infrastructure overlapping with known IRGC-linked clusters; operational tempo consistent with Iranian working patterns; and Mandiant's independent assessment of IRGC nexus. The group's expansion into

Western Europe in September 2025 targeting NATO aerospace contractors further supports state direction rather than criminal motivation.

Alternative hypothesis: UNC1549 could be a contractor group working on behalf of IRGC rather than organic IRGC personnel. This would not materially change the threat assessment but could affect disruption/attribution options. Assessed plausible but not determinative.

Evidence Type	Detail	Confidence
Targeting alignment	Exclusively aerospace/defence; victims aligned with Israeli and Gulf military intelligence priorities	High
Infrastructure overlap	Shared hosting and registration patterns with known IRGC clusters	Medium
Geographic expansion	Expansion to NATO Europe (Sep 2025) consistent with post-Ukraine Iranian intelligence priorities	Medium
Custom toolchain	Investment in bespoke tools (MiniJunk, TRUSTRAP, MiniBrowse) indicates state-level resourcing	High
Mandiant assessment	Mandiant independently assesses IRGC nexus in their UNC1549 reporting	High

4. Background & History

UNC1549 emerged as a distinct tracked cluster in June 2022 and has rapidly become one of the most active Iranian threats to the aerospace and defence sector globally. The group's defining innovation is a vertically integrated fake recruitment infrastructure: it builds convincing career sites impersonating Boeing, Airbus, Teledyne FLIR, and other major aerospace brands, then delivers both credential harvesting and malware through these portals.

A critical operational shift occurred in late 2023 when UNC1549 pivoted to exploiting trusted third-party and supply chain relationships — compromising contractors and partners to reach primary defence targets without directly attacking hardened primes. By October 2024, Mandiant documented the group running highly tailored intelligence-gathering operations across the worldwide aerospace supply chain. September 2025 brought geographic expansion into Western Europe, specifically targeting NATO aerospace contractors in Portugal, Sweden, and Denmark.

Date	Event	Significance
Jun 2022	First confirmed UNC1549 activity	Mandiant begins tracking; Middle East aerospace/defence targeting
2023 Q4	Supply chain pivot	Shift to compromising

		contractors to reach defence primes
Feb 2024	"When Cats Fly" Mandiant disclosure	Custom toolchain publicly documented; Israel/Middle East confirmed
Oct 2024	Worldwide supply chain operations	Tailored global intelligence-gathering campaigns documented
Sep 2025	Western Europe expansion	Portugal, Sweden, Denmark targeted — NATO aerospace contractors
Oct 2025	Sustained campaign confirmed	Multi-year campaign active through at least Q3 2025

5. Objectives & Targeting

UNC1549's primary intelligence requirement is visibility into Israeli and Gulf state military aviation and aerospace capabilities — systems, specifications, programme timelines, and personnel. Secondary targeting covers any NATO aerospace contractor with visibility into Western military aviation. The group's supply chain focus means that even organisations without direct defence contracts are at risk if they supply components, software, or services to primary targets.

Attribute	Detail
Organisation Size	All sizes — supply chain targeting means SME aerospace suppliers are in scope
Technology Environment	Windows-centric; cloud environments (Azure AD, O365); web-facing applications
Specific Technologies	Microsoft Exchange (exploitation); Azure AD (credential targeting); browser credential stores
Primary Lures	Fake Boeing, Airbus, Teledyne FLIR career portals; job offer emails/messages
Supply Chain Focus	Explicitly targets contractors/suppliers to reach defence prime targets

6. TTPs — MITRE ATT&CK

Tactic	Technique	Observed	Frequency	Notes
Resource Dev	T1583.001 — Domains	Yes	High	Fake career sites: Boeing, Airbus, Teledyne FLIR impersonation

Resource Dev	T1583.006 — Web Services (Azure)	Yes	High	Azure infrastructure for C2 and payload hosting
Initial Access	T1566.002 — Spearphishing Link	Yes	High	Links to fake career portal from email/LinkedIn messages
Initial Access	T1199 — Trusted Relationship	Yes	High	Third-party/supply chain compromise to reach primary targets
Initial Access	T1078 — Valid Accounts	Yes	High	Credentials harvested via TRUSTRAP fake auth windows
Execution	T1059.001 — PowerShell	Yes	High	Post-compromise execution and lateral movement
Execution	T1106 — Native API	Yes	Med	MiniJunk uses native Windows APIs for stealthy execution
Persistence	T1053.005 — Scheduled Task	Yes	High	Scheduled tasks for MiniJunk backdoor persistence
Defence Evasion	T1036 — Masquerading	Yes	High	Fake aerospace brand portals; legitimate-looking applications
Defence Evasion	T1078.004 — Cloud Accounts	Yes	High	Azure AD staging; C2 blends with enterprise cloud traffic
Credential Access	T1056.002 — GUI Input Capture	Yes	High	TRUSTRAP: fake Windows/O365 auth window to harvest credentials
Credential Access	T1555.003 — Browser Credentials	Yes	High	MiniBrowse extracts Chrome/Firefox/Edge stored credentials
Collection	T1113 — Screen Capture	Yes	Med	SIGHTGRAB captures periodic

				screenshots
C2	T1102 — Web Services	Yes	High	Azure-hosted C2; traffic blends with enterprise cloud
Lateral Movement	T1199 — Trusted Relationship	Yes	High	Pivots through supply chain to reach defence primes

7. Custom Toolset

Tool	Type	Purpose	Attribution Confidence
MiniJunk	Full backdoor	Primary post-access implant; full RAT capability; native API execution	High — exclusive to UNC1549
MiniBrowse	Credential stealer	Extracts browser-stored credentials (Chrome, Firefox, Edge)	High — exclusive
SIGHTGRAB	Screenshot capture	Periodic desktop screenshots for operator situational awareness	High — exclusive
TRUSTRAP	Fake auth window	Convincing Windows/O365 login prompt to harvest credentials in-session	High — exclusive
TWOSTROKE	Unknown capability	Additional tool; limited public detail	High — custom
DEEPROOT	Unknown capability	Additional tool; limited public detail	High — custom

8. Detection

8.1 Sigma — TRUSTRAP Fake Auth Window

```

title: UNC1549 — Productivity App Spawning Credential-Capture Child Process (TRUSTRAP)
id: b1e2f3a4-c5d6-7890-bcde-f34567890003
status: experimental
description: Office/PDF app spawning scripting engine — consistent with TRUSTRAP delivery
references:

```

```
- https://cloud.google.com/blog/topics/threat-intelligence/suspected-iranian-unc1549-targets-israel-middle-east
author: Lost Boys Cyber | date: 2026-05-29
tags:
  - attack.credential_access
  - attack.t1056.002
logsource:
  category: process_creation
  product: windows
detection:
  selection:
    ParentImage|endswith:
      - "\winword.exe"
      - "\excel.exe"
      - "\acrord32.exe"
      - "\FoxitReader.exe"
    Image|endswith:
      - "\mshta.exe"
      - "\wscript.exe"
      - "\cscript.exe"
      - "\powershell.exe"
  condition: selection
falsepositives:
  - Macros in legitimate documents; baseline environment before deploying
level: high
```

8.2 Sigma — Browser Credential Store Access by Non-Browser Process

```
title: UNC1549 — MiniBrowse Pattern: Non-Browser Accessing Browser Credential Store
id: b2e3f4a5-c6d7-8901-cdef-345678900004
status: experimental
description: Non-browser process reading Chrome/Firefox/Edge Login Data — MiniBrowse pattern
tags:
  - attack.credential_access
  - attack.t1555.003
logsource:
  category: file_access
  product: windows
detection:
  selection:
    TargetFilename|contains:
      - "Chrome\User Data\Default\Login Data"
      - "Firefox\Profiles"
      - "Edge\User Data\Default\Login Data"
  filter:
    Image|endswith:
      - "\chrome.exe"
      - "\firefox.exe"
      - "\msedge.exe"
      - "\GoogleUpdate.exe"
  condition: selection and not filter
falsepositives:
  - Enterprise password managers; validate Image path
level: high
```

8.3 KQL — Azure Guest/External Anomalous File Access

```
// UNC1549: Supply chain pivot — unusual guest account file activity
CloudAppEvents
| where Timestamp > ago(90d)
| where AccountType == "Guest"
```

```
| where ActionType in ("FileDownloaded","FileUploaded","FileSyncDownloadedFull")
| summarize Count=count(), Files=make_set(DestinationFileName, 20)
|   by AccountDisplayName, IPAddress, CountryCode
| where Count > 20
| order by Count desc
```

9. Threat Hunting

9.1 Hunt: Fake Recruitment Domain DNS Lookups

Hypothesis: If UNC1549 targeted employees via fake career portals, DNS queries to impersonation domains (boeing-, airbus-, teledyne-themed on unusual TLDs) will appear in DNS logs.

```
// Proxy/DNS – fake aerospace career domain lookups
DeviceNetworkEvents
| where Timestamp > ago(90d)
| where RemoteUrl has_any
("boeing","airbus","teledyne","flir","lockheed","northrop")
| where RemoteUrl !endswith "boeing.com"
|   and RemoteUrl !endswith "airbus.com"
|   and RemoteUrl !endswith "teledyne.com"
| project Timestamp, DeviceName, AccountName, RemoteUrl, RemoteIP
| order by Timestamp desc
```

9.2 Hunt: New Azure App Registrations (MiniJunk C2 Persistence)

Hypothesis: MiniJunk uses Azure cloud services for C2; operators may register new Azure apps for persistent access.

```
AuditLogs
| where TimeGenerated > ago(90d)
| where OperationName in ("Add application","Update application","Add service principal")
| where InitiatedBy.user.userPrincipalName !in~ (known_admin_list)
| project TimeGenerated, OperationName,
|   Initiator=tostring(InitiatedBy.user.userPrincipalName),
|   AppName=tostring(TargetResources[0].displayName)
| order by TimeGenerated desc
```

10. Recommended Defensive Posture

- Enforce phishing-resistant MFA (FIDO2) on all accounts — TRUSTRAP can capture TOTP codes; hardware keys are the only MFA class that defeats in-session credential capture.
- Harden third-party and supply chain access — UNC1549 pivots through contractors; enforce least-privilege on all guest/external Azure AD accounts and review API integrations quarterly.
- Block job-themed phishing domains at DNS/proxy — configure email and web gateways to flag messages referencing fake aerospace career portals; monitor DNS for Boeing/Airbus impersonation domains.
- Monitor Azure App registrations and OAuth consent — UNC1549 uses Azure for persistence; alert on new app registrations with broad permission scopes from non-administrative accounts.
- Deploy MiniBrowse detection — non-browser process access to Chrome/Firefox/Edge Login Data files is a high-fidelity indicator; implement file access auditing on browser credential store paths.

- Assess supply chain partners — any SME in your aerospace supply chain that lacks MFA on O365 is a viable UNC1549 entry point into your network; include key suppliers in threat awareness briefings.

11. References

- [1] Mandiant/Google Cloud. "When Cats Fly: UNC1549." <https://cloud.google.com/blog/topics/threat-intelligence/suspected-iranian-unc1549-targets-israel-middle-east>
- [2] Mandiant/Google Cloud. "UNC1549 TTPs Analysis." <https://cloud.google.com/blog/topics/threat-intelligence/analysis-of-unc1549-ttps-targeting-aerospace-defense>
- [3] Picus Security. "UNC1549 TTPs." <https://www.picussecurity.com/threat-database/unc1549-ttps-iranian-apt-targeting-aerospace-and-defense>
- [4] The Hacker News. "Iran-linked UNC1549." <https://thehackernews.com/2024/02/iran-linked-unc1549-hackers-target.html>
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